

Portfolio Adaptation Across the Wealth Distribution: Evidence from U.S. DFA Local Projections

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Abstract

This thesis...

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Chapter 1

Introduction

1.1 Research Question

Wealth groups differ substantially in their exposure to financial markets, their willingness to take risk, and their ability to adapt portfolios when asset prices move, with wealthier households typically holding more risky assets and facing lower constraints. While much is known about how individual investors behave in micro-level data, far less is understood about how groups of households across the wealth distribution adapt their portfolios in real time when equity values change, particularly in a setting where balance-sheet components can be tracked consistently at a medium frequency. This gap is important because differences in rebalancing behavior shape how valuation shocks propagate through balance sheets and influence long-run, absolute wealth accumulation despite ongoing wealth inequality. The research question is therefore whether wealthier groups rebalance their equity positions more promptly when confronted with unforeseen valuation movements. Using the U.S. Distributional Financial Accounts (DFA), which provides quarterly balance-sheet data for distinct wealth groups, the analysis examines how they respond to group-specific valuation shocks particularly in corporate equities and mutual fund holdings, relying on local projections to capture flexible, horizon-specific adjustments.

1.2 Relevant Literature

Research on household portfolios consistently shows that wealth groups differ sharply in both their asset composition and their exposure to valuation shocks. Using long-run SCF microdata, [Bricker et al. \(2019\)](#) document that the typical U.S. household holds a balance sheet dominated by housing and illiquid assets, while only the upper tail of the wealth distribution holds substantial business equity and diversified financial market assets. As they emphasize, “most families hold a relatively small share of assets in financial markets and instead hold primarily housing assets,” whereas the top of the distribution is characterized by large positions in private businesses and equities. These structural differences imply that unexpected changes in asset values propagate very differently across the wealth distribution, shaping both the magnitude of valuation shocks households experience and the channels through which they can adapt.

A second strand of the literature focuses on the distinction between passive drift and active rebalancing. [Calvet et al. \(2008\)](#), using comprehensive administrative data from Sweden, show that

households do not simply absorb asset price movements mechanically. Instead, they actively rebalance their portfolios in response to idiosyncratic fluctuations in risky asset values. Their decomposition reveals that active rebalancing “on average offset[s] about one half of idiosyncratic passive variations” in risky asset shares. Importantly, the intensity of rebalancing varies systematically across households. Wealthier, better educated, and better diversified households adjust more aggressively toward their target allocations, while less sophisticated households exhibit inertia, limited monitoring, and occasional procyclical behavior. These findings provide the microeconomic foundation for distinguishing mechanical valuation effects from deliberate portfolio adjustments and highlight that rebalancing is a behavioral margin that differs across the wealth distribution.

A complementary empirical literature examines heterogeneity in returns to wealth. [Fagereng et al. \(2016\)](#), using two decades of Norwegian administrative data, document substantial dispersion in returns even within asset classes. They find a spread of roughly five hundred basis points between the tenth and ninetieth percentiles of the return distribution and show that returns contain a large permanent component that “accounts for 60 percent of the explained variation.” They also show that returns rise with wealth, stating that “returns are positively correlated with wealth.” These patterns imply that wealthier households not only hold different portfolios but also experience different return processes, with larger and more persistent valuation shocks. This heterogeneity in return dynamics reinforces the need to study portfolio adjustments separately across wealth groups and suggests that wealthier households may have both stronger incentives and greater capacity to rebalance after shocks.

Recent evidence from the United States further highlights the importance of wealth dependent trading behavior. [Gabaix et al. \(2023\)](#), using high frequency security level data, show that less affluent households tend to sell equities during market downturns, while ultra high net worth households tend to buy and thereby act in a countercyclical manner. Despite these contrasting directions, the overall elasticity of flows is small. The authors emphasize that the sensitivity of flows to returns is “generally quite small,” indicating that even wealthy households exhibit substantial inertia. Their distinction between an investor’s own return relative to the market and the market return itself parallels the distinction between active and passive portfolio changes and underscores the importance of measuring behavioral responses relative to a passive benchmark. Their findings also suggest that the direction of rebalancing may differ across wealth groups even when the magnitudes are modest, a pattern that flexible empirical methods are well suited to detect.

Before examining the empirical implications of heterogeneous portfolio behavior, it is useful to situate these patterns within the broader conceptual framework of household finance. [Campbell \(2006\)](#) emphasizes that households operate in an environment shaped by fixed participation costs, limited attention, uninsurable income risk, borrowing constraints, and complex financial products. These frictions generate systematic deviations from textbook portfolio theory and help explain why households with different levels of wealth and sophistication behave so differently when asset values move. Campbell notes that while many households manage their finances reasonably effectively, “a minority make significant mistakes,” and that this minority tends to be poorer and less educated. He also highlights that certain financial products involve “a cross subsidy from naive households to sophisticated households,” which can discourage the development of products that would support more effective financial decision making. This conceptual perspective provides a foundation for understanding why wealthier households, who face fewer constraints and possess greater financial sophistication, tend

to adjust their portfolios more actively after valuation shocks, while lower wealth households often display inertia, liquidity driven adjustments, or procyclical responses.

These empirical and conceptual insights are central to modern heterogeneous agent macroeconomics, where models of wealth inequality rely on persistent differences in returns, portfolio composition, and exposure to idiosyncratic capital income risk. In the canonical framework of [Benhabib and Bisin \(2009\)](#), idiosyncratic and persistent heterogeneity in returns to wealth generates Pareto tailed wealth distributions, with the right tail shaped primarily by capital income risk rather than labor income risk. [Gabaix et al. \(2015\)](#) further show that to explain the dynamics of top inequality, particularly the rapid rise observed in recent decades, models must incorporate size dependent returns or heterogeneity in mean growth rates. These mechanisms closely mirror the empirical evidence on wealth dependent returns and rebalancing behavior. [Nirei and Aoki \(2016\)](#) demonstrate that entrepreneurial productivity risk, combined with borrowing constraints and incomplete markets, naturally produces Pareto tailed income and wealth distributions within a Bewley type general equilibrium model. Taken together, this body of work shows that heterogeneity in returns, portfolio composition, and rebalancing behavior is not merely a microeconomic curiosity but a central mechanism in macroeconomic theories of inequality. The empirical findings reviewed above therefore provide the micro foundations needed to discipline heterogeneous agent models and motivate examining how different segments of the wealth distribution adapt their portfolio shares after valuation shocks.

Chapter 2

Methodology

2.1 Data

The analysis uses the Distributional Financial Accounts (DFA) introduced by [Batty et al. \(2022\)](#), a quarterly dataset beginning in 1989Q3 that integrates the national-accounting structure of the Financial Accounts with the micro detail of the Survey of Consumer Finances (SCF). The DFA reports balance-sheet components for wealth groups of bottom 50%, 50–90%, 90–99%, 99–99.9%, and top 0.1%, and provides both levels and shares of major asset and liability categories reconciled to the Financial Accounts framework.

These data include group-level equity holdings, equity portfolio shares, fixed income, cash, private businesses, nonfinancial assets, and liabilities, which form the core variables in the empirical analysis. Because the DFA does not report market participation rates, changes in portfolio shares must be interpreted as group-level adjustments that combine both intensive-margin trading and extensive-margin entry or exit.

The project focuses on structural behavioral differences across groups, not on tracking the same individuals. Each wealth group represents a distinct economic environment, so individuals who move between groups are entering a different behavioral regime rather than contaminating the original one. Moreover, the local-projection framework estimates short-horizon responses (within 1 to 2 years), over which mobility across broad wealth quantiles is empirically limited. For these reasons, the DFA’s group-level aggregates remain the appropriate object.

The category “corporate equities and mutual fund shares” excludes private equity and private business assets. Because this category reflects public-market equity positions, the most coherent way to construct a passive benchmark is to apply a broad public-market return, even though wealth groups differ in how diversified their holdings are within this category. Methodological implications of this choice are discussed in detail in the subsequent section.

Before turning to the empirical strategy, the dataset will be explored descriptively. The DFA also provides distributions by education and age in addition to wealth percentiles. These alternative groupings offer additional context for understanding how portfolio composition and exposure to valuation shocks vary, possibly in light of financial literacy and life cycle aspects.

2.2 Hypothesis

The literature highlights systematic differences across the wealth distribution in shaping portfolios, inducing the following hypothesis.

On one hand, wealthier households actively rebalance their equity portfolios after valuation shocks. Groups with higher wealth, particularly those holding substantial corporate equities and mutual fund shares, are expected to adapt their portfolio shares more rapidly due to greater diversification, higher financial sophistication, and lower liquidity constraints.

On the other hand, lower-wealth households exhibit limited active rebalancing, with adjustments reflected mainly in passive drift and liquidity movements. For the bottom half of the distribution, equity positions are quite small relative to housing, implying that changes in equity shares largely reflect mechanical price movements. At the same time, liquidity constraints and precautionary motives suggest that valuation shocks may influence shares in cash, making liquidity adaptations an important complementary margin for these households.

2.3 Empirical Frameworks

2.3.1 Identify Group-Specific Valuation Shocks

Unexpected changes in equity values are identified using residuals from a predictive equation for the corresponding wealth accumulation process:

$$\Delta \ln L_{g,t} = \alpha_g + \phi_g \Delta \ln L_{g,t-1} (+\beta_g \bar{r}_t) + \varepsilon_{g,t},$$

where $L_{g,t}$ denotes the group's equity holding levels and $\bar{r}_t$ is the overall market excess return adjusted by the risk-free rate. The residual $\varepsilon_{g,t}$ is interpreted as a group-specific valuation shock, capturing unexpected changes in equity values after accounting for market movements and past dynamics. This specification is intended to isolate “local” shocks that are specific to each group's equity portfolio, net of aggregate market fluctuations.

To assess the sensitivity of the results to the shock definition, the analysis also considers a specification that omits the market-return term. Removing $\bar{r}_t$ yields a residual that reflects a broader equity-wealth shock, combining both market-driven and idiosyncratic components. This backup, simplest interpretation treats any unexpected change in equity wealth as the “global” disturbance.

2.3.2 Decompose Rebalancing Behavior

An imperative piece of the literature is the distinction between passive drift i.e., mechanical changes in portfolio shares due to asset-price movements, and active rebalancing, which reflects deliberate trading behavior. To incorporate this distinction, a passive benchmark for the equity share is constructed by revaluing equity holdings using average market returns:

$$L_{g,t+h}^P = L_{g,t} \prod_{j=1}^h (1 + \bar{r}_{t+j}).$$

The corresponding passive equity share is obtained by dividing this by the actual total assets observed in the DFA:

$$S_{g,t+h}^P = \frac{L_{g,t+h}^P}{A_{g,t+h}}.$$

This approach defines passivity relative to the realized evolution of the group's balance sheet, avoiding the pain to model group-specific total portfolio returns. Even though wealth groups differ in diversification and actual returns, the passive benchmark uses a common market return because it represents the mechanical effect of broad market movements. Subtracting this benchmark isolates the behavioral component of portfolio adaptation:

$$S_{g,t+h}^A = S_{g,t+h} - S_{g,t+h}^P.$$

This active rebalancing component measures how a group's equity share changes relative to the passive benchmark implied by market returns. A positive value indicates that the group increased its equity share more than passive drift would predict, while a negative value indicates active selling or a shift toward safer or more liquid assets. This interpretation holds regardless of whether average market returns are positive or negative: when returns are positive, the passive share rises mechanically, so a positive active component reflects reinforcing the boom; when returns are negative, the passive share falls mechanically, so a positive active component reflects cushioning the decline or buying into a downturn. In all cases, the sign of the active component cleanly captures the direction of deliberate portfolio adaptation relative to what would have happened had the group not traded at all as a counterfactual.

2.3.3 Allow Flexibility in Passive Drifts

To allow the passive benchmark to reflect systematic differences in public-equity composition across wealth groups, the analysis constructs an implied return series from the DFA's equity-level data. For each group g , the log-difference of equity holdings defines the constructed return:

$$\hat{r}_{g,t} = \ln L_{g,t} - \ln L_{g,t-1},$$

where $\hat{r}_{g,t}$ is an inferred return rather than an observed market series. Because DFA equity levels combine valuation changes with net flows, this transformation provides only an approximate measure of the underlying return process, but it offers a consistent way to extract the component of equity wealth that co-moves with aggregate market movements.

Group-specific exposure to aggregate equity fluctuations is then estimated by regressing the constructed return on the aggregate market return:

$$\hat{r}_{g,t} = \varphi_g + \psi_g \bar{r}_t + \eta_{g,t}.$$

The coefficient ψ_g captures the extent to which each group's public-equity holdings co-move with broad market movements, while φ_g absorbs average drift and $\eta_{g,t}$ captures idiosyncratic variation arising from flows, revaluations, and other group-specific factors.

The estimated exposure parameter is incorporated into a group-specific passive benchmark:

$$L_{g,t+h}^P = L_{g,t} \prod_{j=1}^h (1 + \psi_g \bar{r}_{t+j}) ,$$

which preserves the interpretation of a purely mechanical evolution of public-equity wealth in the absence of active trading, while allowing the magnitude of passive drift to differ across groups in a manner consistent with their estimated exposure to aggregate market movements.

The passive equity share and the corresponding active rebalancing component are then computed using this adjusted benchmark, ensuring that all subsequent portfolio-adaptation measures reflect the group-specific sensitivity of public-equity wealth to aggregate market movements.

2.3.4 Core: Local Projections

Portfolio responses to valuation shocks are estimated using the basic local-projection framework of [Jordà and Taylor \(2024\)](#), applied separately to each wealth group. For each horizon h , the following specification is estimated:

$$S_{g,t+h} = \gamma_{g,h} + \delta_{g,h} \varepsilon_{g,t} + \rho'_{g,h} X_{g,t} + u_{g,t+h} ,$$

where $S_{g,t}$ is the equity portfolio share, $\varepsilon_{g,t}$ is the group-specific valuation shock, and includes its own lagged portfolio share ($S_{g,t-1}$) as well as balance-sheet controls in levels that mechanically affect portfolio shares in the short run (i.e., fixed income, private businesses, real estate, and liabilities).

where $S_{g,t}$ is the equity portfolio share, $\varepsilon_{g,t}$ is the group specific valuation shock, and $X_{g,t}$ includes its own lagged portfolio share ($S_{g,t-1}$) as well as balance sheet controls in levels that mechanically affect portfolio shares in the short run (fixed income, private businesses, real estate, and liabilities). Although some balance sheet components such as pensions and miscellaneous assets are not modeled explicitly, their slow moving nature and their presence in the portfolio share denominator imply that most of their influence is absorbed mechanically, while the included controls capture the major short run rebalancing channels that could otherwise confound the estimated responses.

To assess whether valuation shocks also affect liquid asset positions, the same local projection structure is estimated with the cash share as the outcome variable. Cash, defined as deposits and money market fund shares, serves as the primary liquidity buffer for households, particularly for lower wealth groups. Incorporating this margin within the same empirical framework provides a concise way to detect whether valuation shocks lead to precautionary liquidity accumulation or shifts between risky and safe assets, without requiring a separate modeling structure.

Together, these specifications characterize how households adjust both their risky asset exposure and their liquidity positions in response to valuation shocks. The key step is then to isolate the behavioral component of equity adaptation by re-estimating the local projection using the active rebalancing measure $S_{g,t+h}^A$ as the dependent variable. This final specification removes the mechanical influence of market driven valuation changes and focuses exclusively on deliberate portfolio adjustments across wealth groups, which is central to the empirical contribution of the analysis.

2.3.5 Asymmetric Shock Responses

To capture whether households react differently to gains and losses, the local-projection specification is augmented to allow separate coefficients for positive and negative valuation movements:

$$S_{g,t+h} = \gamma_{g,h} + \delta_{g,h}^+ \varepsilon_{g,t}^+ + \delta_{g,h}^- \varepsilon_{g,t}^- + \rho'_{g,h} X_{g,t} + u_{g,t+h},$$

where the shock components are defined as

$$\varepsilon_{g,t}^+ = \max(\varepsilon_{g,t}, 0), \quad \varepsilon_{g,t}^- = \min(\varepsilon_{g,t}, 0).$$

This formulation isolates whether negative shocks trigger stronger adjustments than positive shocks of the same magnitude, consistent with loss aversion. $|\delta_{g,h}^-| > |\delta_{g,h}^+|$ would indicate that groups reduce risk exposure more forcefully after losses, while similar coefficients would imply symmetric adjustment behavior.

2.3.6 Extended: Time-Varying Local Projections

Asset composition has evolved substantially over the sample period, alongside experiencing major financial disruptions. To capture potential shifts in rebalancing behavior, split-sample LPs are estimated:

$$S_{g,t+h} = \gamma_{g,h}^{(\tau)} + \delta_{g,h}^{(\tau)} \varepsilon_{g,t} + \rho_{g,h}^{(\tau)} X_{g,t} + u_{g,t+h}^{(\tau)},$$

where $\tau \in \{\text{pre-crisis, post-crisis}\}$ indexes the two subsamples before and after the 2008 financial crisis. This approach allows the impulse responses to differ flexibly without imposing parametric drift, providing insight into whether rebalancing speeds have changed as household balance-sheet structures evolved.

2.3.7 Robustness: Panel LPs & BVARs

Two additional approaches are used to assess robustness. First, panel LPs introduce group fixed effects, which absorb time-invariant differences in portfolio levels and composition across wealth groups, and time fixed effects, which control for common macroeconomic conditions that affect all groups simultaneously. Second, Bayesian VARs provide a check on whether the joint dynamics of equity holdings and portfolio shares are consistent with the LP results under a richer dynamic structure with shrinkage accounting for heteroskedasticity.

Chapter 3

Results

3.1 Main Findings

Chapter 4

Conclusion

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